Summer 2026 Lab 1: Getting Started with ArcGIS Pro

CMP 4450/6450

May 4, 2026

The Department of Health is concerned with density of health facilities in some of the larger and more sparsely populated counties within the state of Utah. they have asked that we find the three largest counties in the state and determine how many facilities are within these three counties.

DATA TO DOWNLOAD: files to upload to ArcGIS

Overview of Steps: List of everything you'll accomplish in this lab
Learning Objectives: What tools and functions you'll use and learn
Workflow: Process map of beginning to end of lab

LAB BEGINS HERE: Where you will start working on ArcGIS

Video Topic: What actions you will be doing or learning
New Tools: Intro to a new tool and what it does
Steps: Breakdown of how to access and run tools

Tips/Notes: Where to look if something isn't working

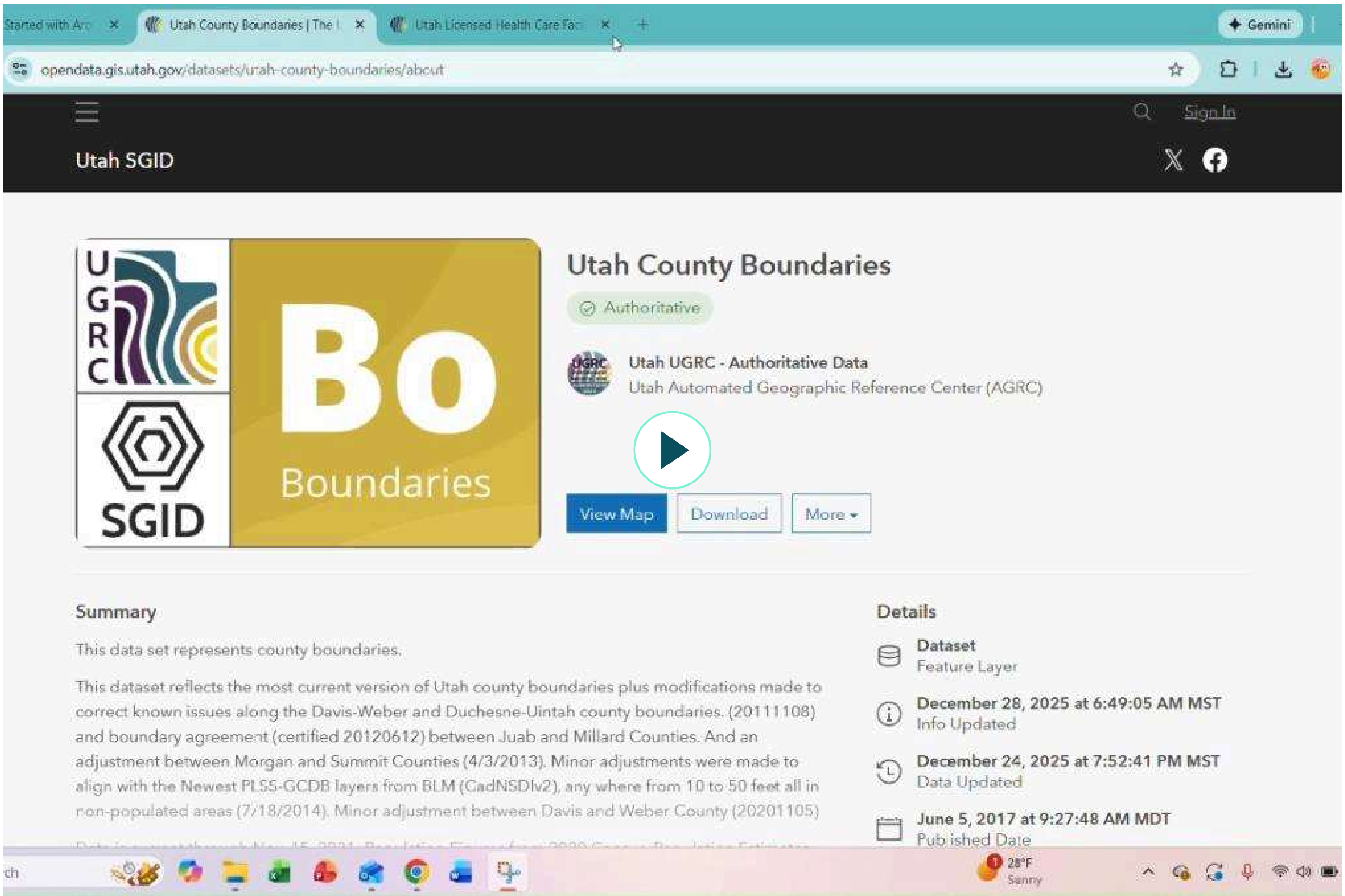
LAB SUBMISSION: What you need to submit to Canvas

Lab Configuration

Each lab will have written instructions over here, and accompanying videos on the right-hand side of the screen. There may be screenshots of tool windows included throughout the written instructions! As you progress through the labs, some videos of simple steps will be replaced by photos. Check out the screenshot to the right (where videos will be) to see how each part of the lab will be labeled and the general order we will follow.

Note: Not every lab will have every part listed to the right! They will all have the parts written in red, blue, and maroon.

Once you familiarize yourself with this order, scroll down to see it in action as you try out your first lab!



DATA TO DOWNLOAD:

Counties shapefile
Utah healthcare facilities

To download both of these, click the link > Download > Shapefile Download.

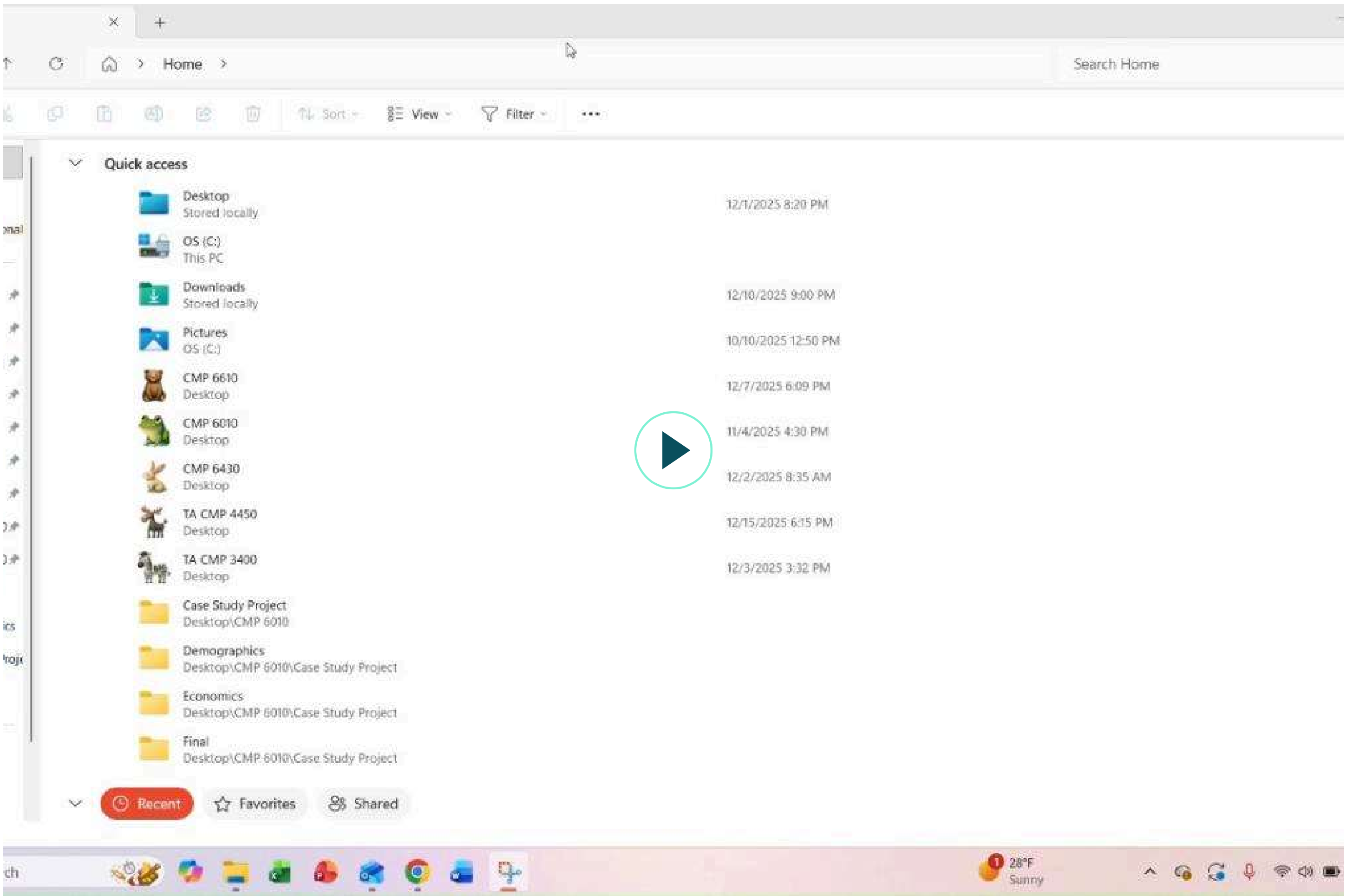
Note: If you download a ZIP file, you will always want to extract the data before you work with it. We will go over this later in the lab.

Learning Objectives

In this lab you will learn the basic components of the ArcGIS interface, as well as how all labs will be formatting throughout the semester.

Overview of Steps

1. Identify the components of the ArcGIS Pro interface (such as homepage, catalog and contents panes, and some of the primary tabs/buttons on the ribbon)
2. Adding layers to the Contents window.
3. Simple Selection.
4. Attribute table.
5. Map tab (Locate, Bookmark, Basemap, Zoom, ...)



LAB BEGINS HERE

Organizing your Files and Folders

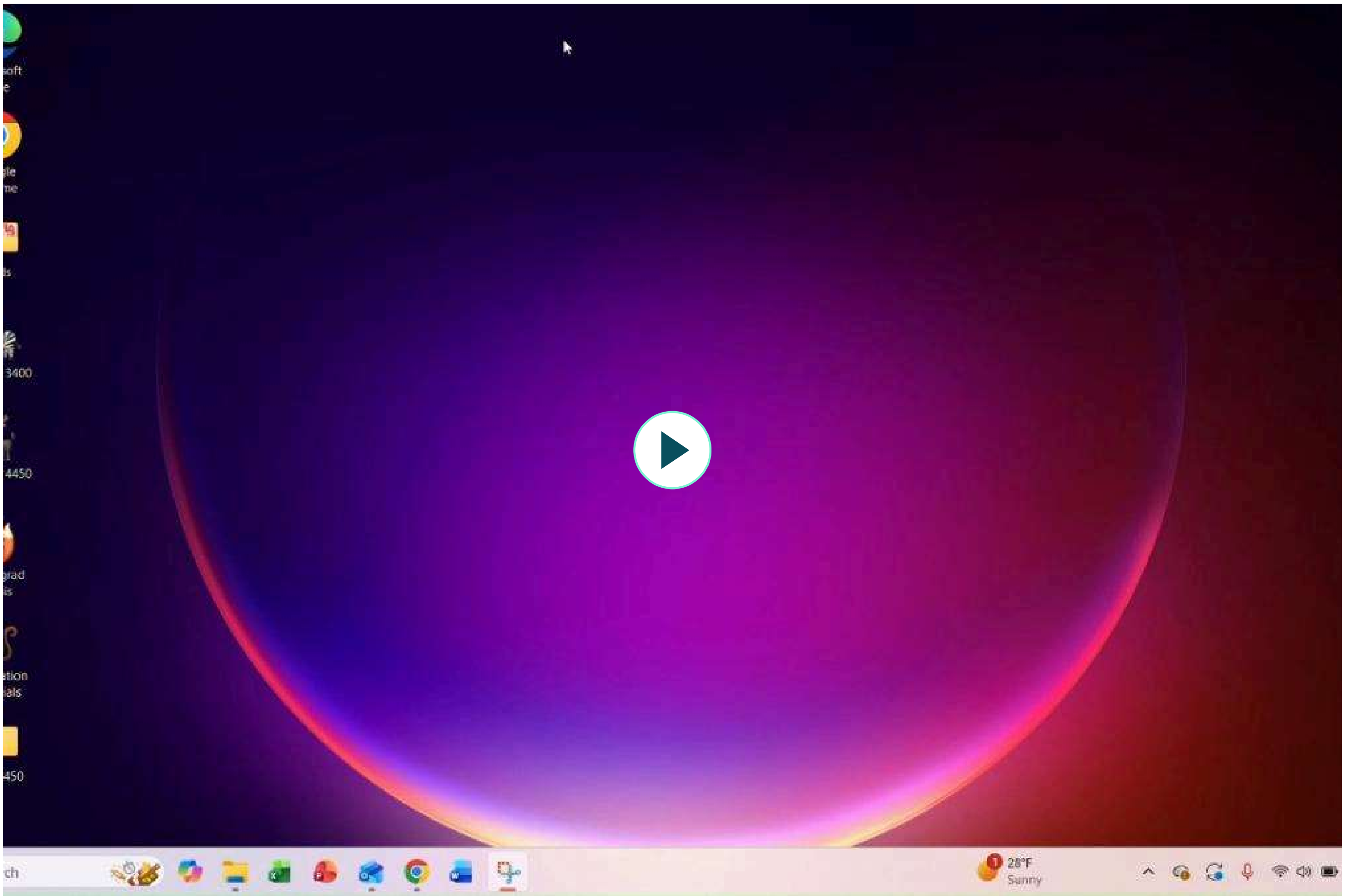
When working in ArcGIS on a project, you often have a lot of data that you are working with and if your folder structure isn't organized things can get messy really fast. Logical Folder structure will help you stay organized with the many layers of data, maps, and exports you will be using throughout the course

(If you never use GIS again, at least you will be well organized in folder structure within other ventures in your personal and professional life)

Steps:

- **Step 1:** Click on your File Explorer
- **Step 2:** Navigate to wherever you want to store your class folder (I use Desktop for all my classes, including this one)
- **Step 2.5:** We HIGHLY recommend, if you are using library PCs or laptops, that you bring a flash drive to store all your files on
- **Step 3:** In the top right corner, click New > Folder
- **Step 4:** Rename your folder to CMP 4450 (or 6450 if you are a graduate student)
- **Step 5:** Open up your new folder, then repeat Step 3. Rename this folder Lab 1
- **Step 6:** Move all your downloaded data to this folder.
- **Step 7:** Extract all the data by right-clicking the file and clicking Extract All.

Note: We HIGHLY recommend, if you are using library PCs or laptops, that you bring a flash drive to store all your files on.



Starting a New Project

In ArcGIS Pro, maps and data are organized in a project. For this class, you will access new projects by clicking Map, and existing projects through the list of Recent Projects or the Open Another Project button.

Steps:

- **Step 1:** Open the software by double clicking ArcGIS Pro
- **Step 2:** Create a new project by clicking Map
- **Step 3:** A window titled New Project should appear. Under Name, title your project Lab 1 Map. Under Location, find the folder you have for this class and your first lab. Click OK.

Navigating ArcGIS Pro

Now that you have created a project, you are greeted by the standard map view of ArcGIS Pro and can get to work. ArcGIS Pro has two base map layers by default: the World Topographic Map and World Hillshade, which you can see in the middle of your workspace. When you open a new project, you start with a global view and can zoom to data that is added.

Steps:

- Step 1: Click on all the tabs along the top Menu bar. Explore some of the features of each button, then click back on Map.
- Step 2: Click on Explore. You are now able to drag your view around on the map and zoom in/out.
- Step 3: Navigate your cursor over to the Contents Pane on the left-hand side of the screen. This is where your layers will show up once you add them. Click on the check box next to World

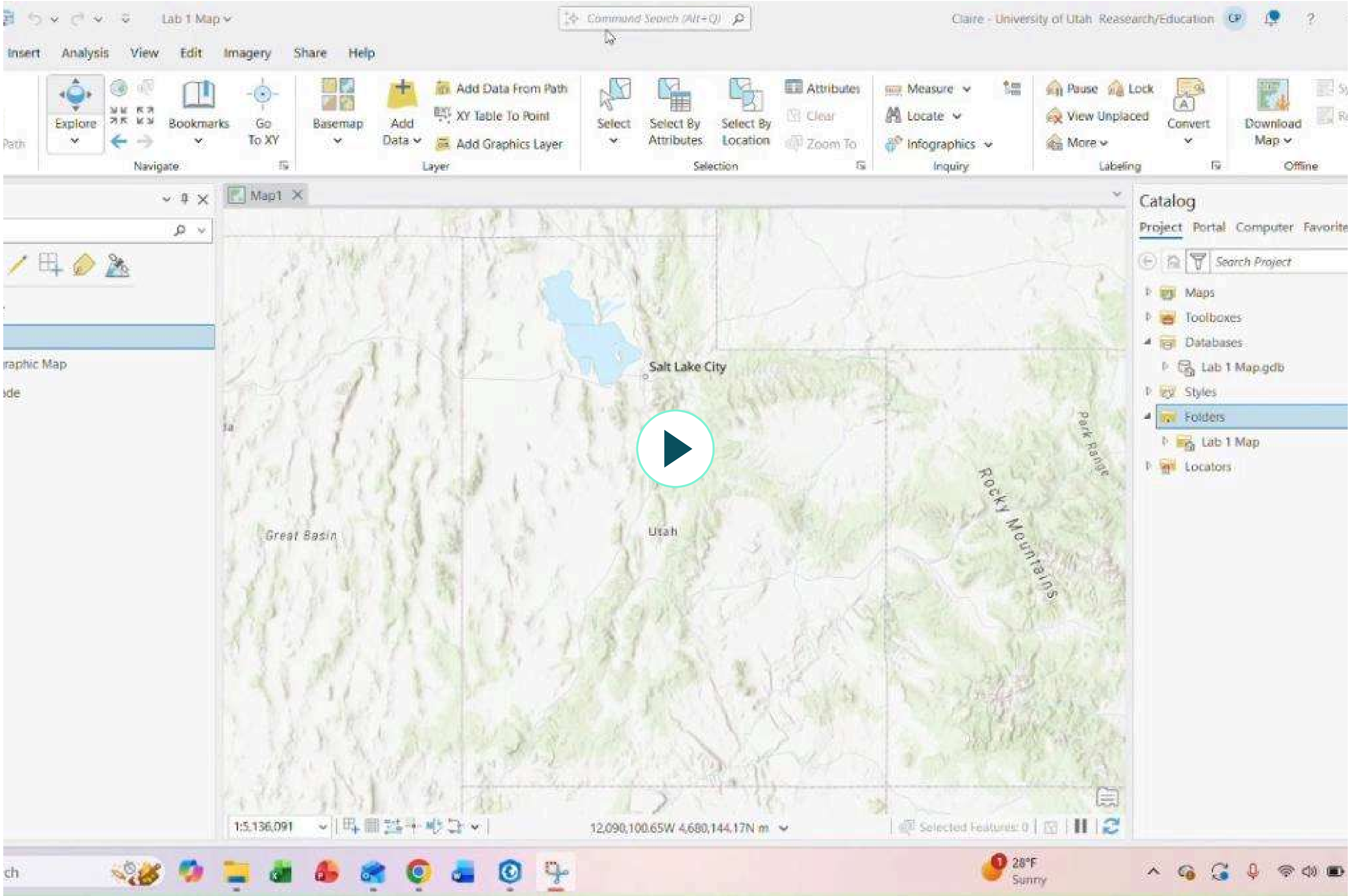
Topographic Map. It is now not visible on your map. Click the check box again to turn visibility back on.

- Step 4: Navigate your cursor to the Catalog Pane on the right-hand side of the screen. Click on the dropdown arrows next to Databases and Folders. You'll be able to see the project database and home folder we made in the previous video.

Note: Sometimes, you may accidentally close your Contents Pane, Catalog Pane, or even your Map! If you do this, click View along the menu ribbon and find whatever it is you need to open up again.

Tip: Esri has many resources that you have access to that are great to have on hand in order to refer to whenever you are stuck. One of the best skills you can walk away with after this course is the ability to find answers to any question you have by knowing where to look. Bookmark these resources and be sure to use them throughout the semester:

https://new-user-learngis.hub.arcgis.com/
https://learn.arcgis.com/en/gallery/
https://www.esri.com/training/catalog/57630435851d31e02a43f007/getting-started-with-arcgis-pro/
https://www.esri.com/training/catalog/5b733d0c8659c25ea7013df9/arcgis-pro-fundamentals/
https://gis.stackexchange.com/



Adding Layers

Our labs will extensively require you to add various datasets to your map in order to complete analysis and create maps. There are a couple ways in which you can add data and we will break them down for you below and with the accompanying video.

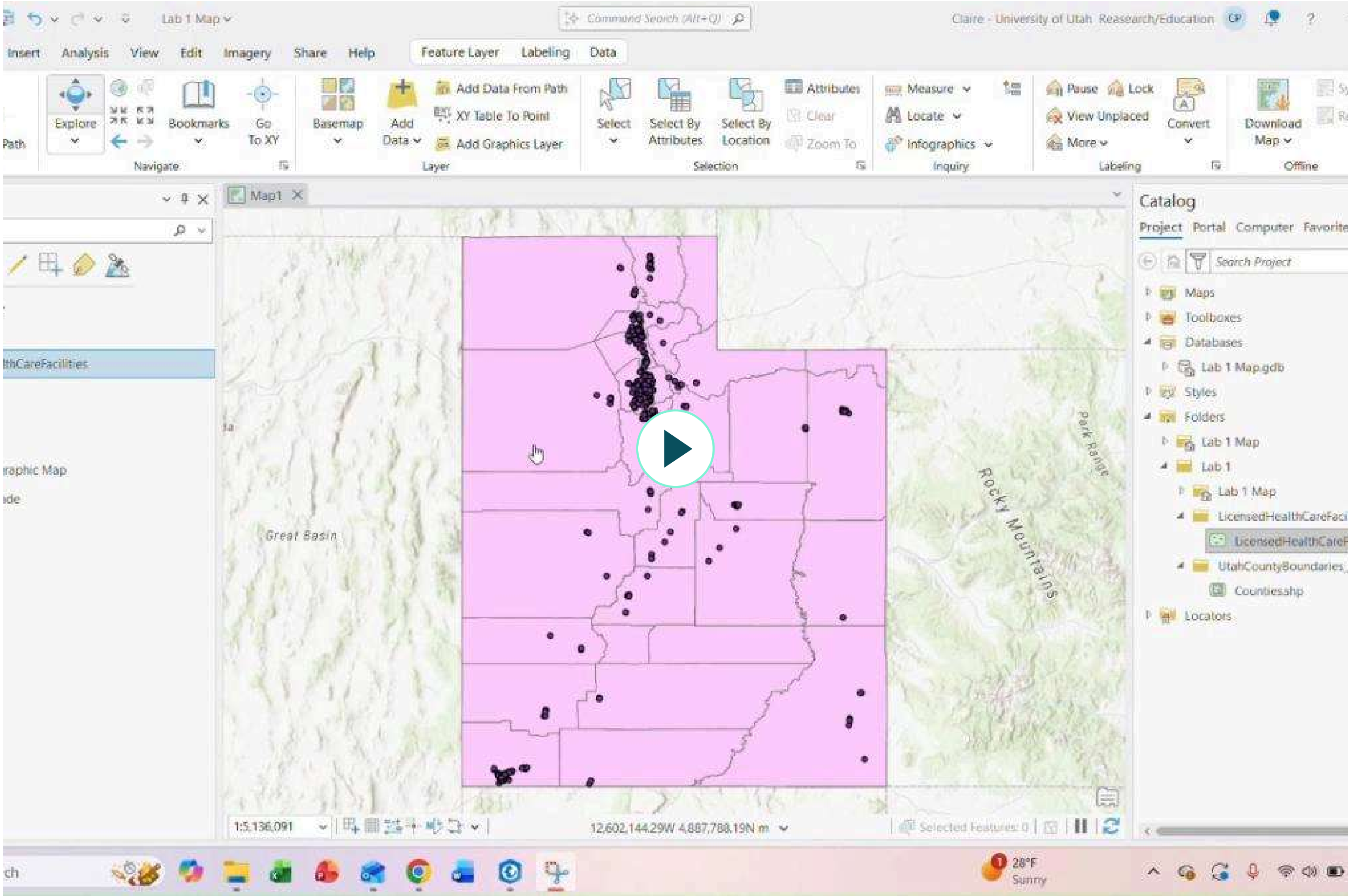
Steps (via Map Tab):

- **Step 1:** On the Menu bar, click Map > Add Data (dropdown arrow) > Browse
- **Step 2:** Navigate to your CMP 4450 Folder > Lab 1 Folder > UtahCountyBoundaries Folder
- **Step 3:** Click Counties.shp, then click Open
- **Step 4:** Repeat Steps 2 and 3, opening the LicensedHealthCareFacilities Folder and Shapefile instead.

Steps (via Folder Connection):

- **Step 1:** On the Catalog Pane, right-click on Folders > Add Folder Connection
- **Step 2:** Navigate to your Lab 1 Folder, click it once and then click OK
- **Step 3:** Click the drop down arrow next to your Lab 1 Folder, then click the drop down next to the UtahCountyBoundaries Folder
- **Step 4:** Right-click Counties.shp, then click Add To Current Map. Alternatively, you can click-and-drag the Shapefile onto the map
- **Step 5:** Repeat steps 3 and 4, this time with the LicensedHealthCareFacilities Folder and Shapefile

Tip: Adding data via Map Tab will be easier for small numbers of data sets. If you have a lot of data to add, it'll be easier to use Folder Connections.



Making a Selection

What is Select?

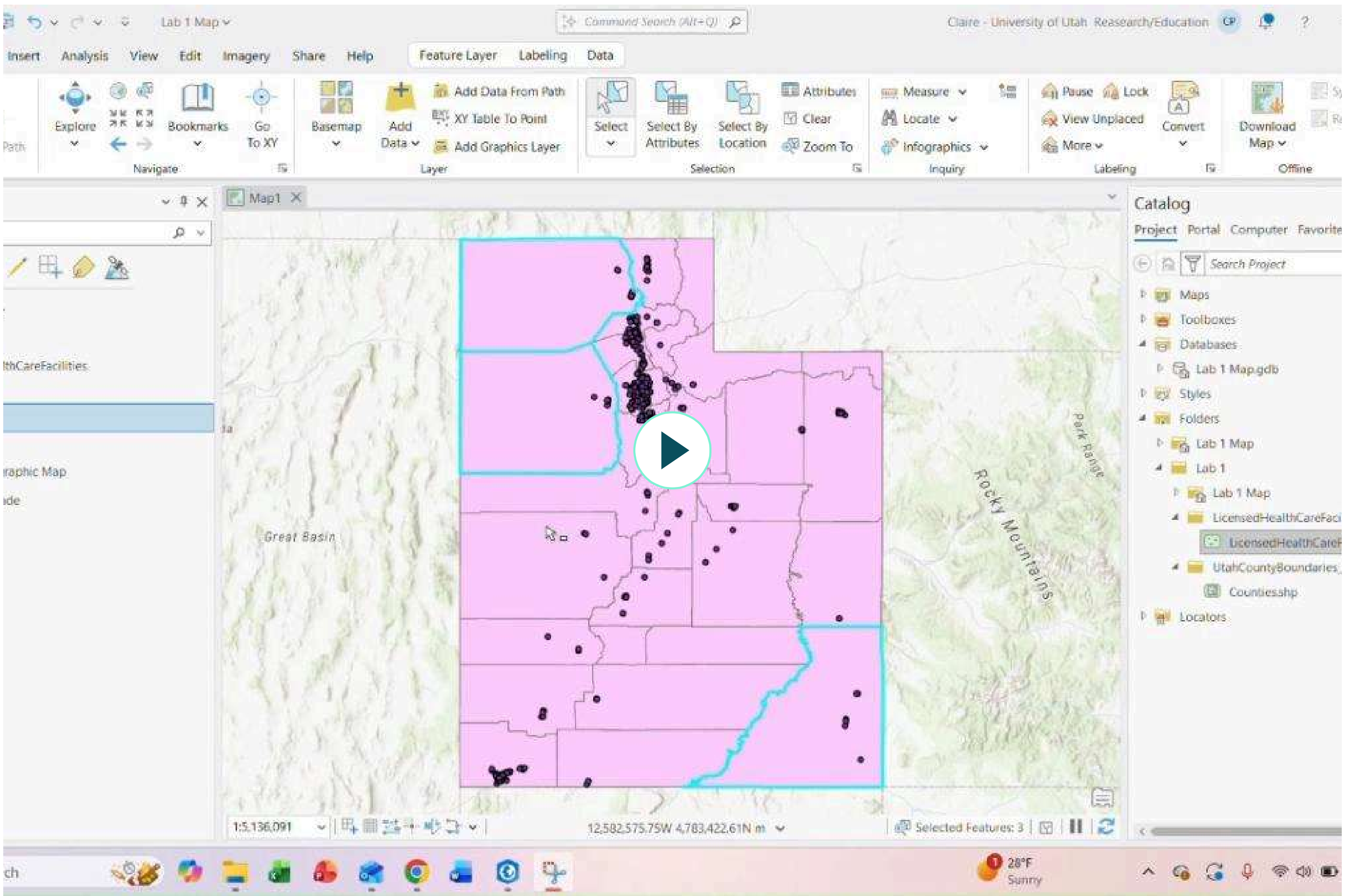
Select allows you to highlight and choose a subset of features on your map to use in subsequent exploration or analysis of your data. There are three main ways of selecting features: simple select, select by attribute, and select by location.

We will be using Simple Select. If you click the Simple Select dropdown arrow, you will notice many tool options: Rectangle (the default), Polygon, Lasso, Circle, Line, and Trace. You will find some of these tools to be more useful than others for different scenarios.

Steps:

- **Step 1:** On the Map Tab, click Select
- **Step 2:** Click on a county of your choice

- **Step 3:** Up near the Select tool, you should see a button called Attributes (this is different than the attribute table from above!). Click that and you will be able to see different characteristics about the county you chose
- **Step 4:** Deselect your selection by clicking Clear at the top of your screen (below Attributes)
- **Step 5:** Hold Shift on your keyboard and select the three counties with the greatest area (Box Elder, Tooele, and San Juan)

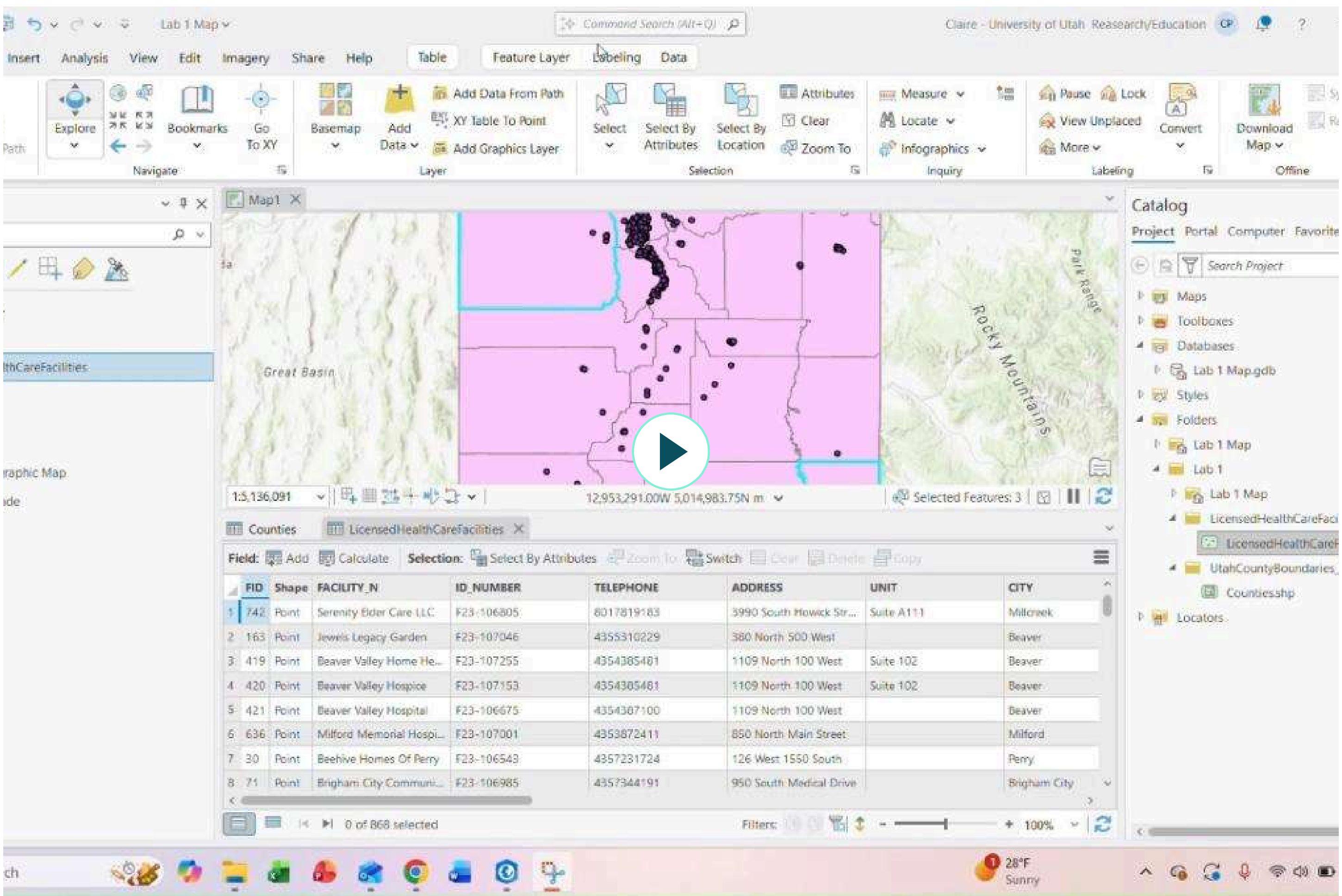


Opening a Layer's Attribute Table

All layers have attribute tables, where nonspatial information about geographic features are contained.

Steps:

- **Step 1:** On your Contents Pane, right-click LicensedHealthCareFacilities
- **Step 2:** Click on Attribute Table
- **Step 3:** Navigate through the Attribute Table until you find the County column
- **Step 4:** Right-click County, then choose Sort By Ascending
- **Step 5:** Manually count the number of facilities in each of the three counties with the greatest areas. Make note of the counts
- **Step 6:** Close the attribute table



LAB SUBMISSION

Congratulations on finishing your first lab! Please screenshot (windows+shift+s) the map you have made with all layers turned on and the three counties with the greatest areas selected. Turn in this screenshot as well as a text submission/word file listing the three counties and their respective healthcare facility counts.